

**Computer Vision for Assistive Technologies: Enhancing
Environmental Perception for the Visually Impaired**

DISSERTATION

**Submitted in Partial Fulfillment of
the Requirements for
the Degree of**

DOCTOR OF PHILOSOPHY (Computer Science)

at the

**NEW YORK UNIVERSITY
TANDON SCHOOL OF ENGINEERING**

by

Yu Hao

May 2025

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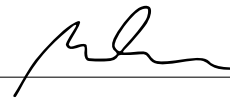
**NEW YORK UNIVERSITY
TANDON SCHOOL OF ENGINEERING**

by

Yu Hao

May 2025

Approved:



Department Head Signature

May 8, 2025

Date

Approved by the Guidance Committee:

Major: PhD Computer Science

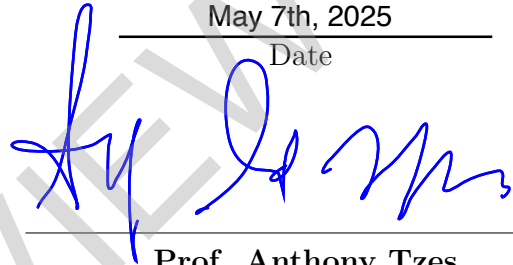


Prof. Yi Fang

Associate Professor of Electrical and Computer Engineering
NYU Tandon School of Engineering

May 7th, 2025

Date

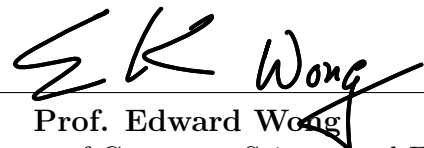


Prof. Anthony Tzes

Professor of Electrical and Computer Engineering
NYU Tandon School of Engineering

May 8th, 2025

Date

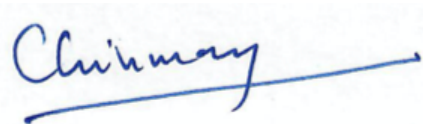


Prof. Edward Wong

Associate Professor of Computer Science and Engineering
NYU Tandon School of Engineering

5/7/2025

Date



Prof. Chinmay Hegde

Associate Professor of Computer Science and Engineering
NYU Tandon School of Engineering

May 8th, 2025

Date

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Vita

Yu Hao received his B.Eng. degree in Computer Science and Technology from Beijing University of Technology, Beijing, China, in 2019. He obtained his M.S. degree in Computer Science from the Tandon School of Engineering, New York University, USA, in 2021. During his master's studies, he developed a strong interest in deep learning, artificial intelligence, and assistive technology, which motivated him to pursue a Ph.D. degree.

In September 2021, Yu Hao enrolled in the Ph.D. program in the Department of Computer Science and Engineering at the Tandon School of Engineering, New York University. Under the supervision of Professor Yi Fang, he has been working at the NYU Embodied AI and Robotics Lab, focusing on deep learning for assistive technology. His research explores vision-based assistive systems for people with blindness and low vision, leveraging deep learning models to enhance navigation, object recognition, and spatial cognition.

Yu Hao has published a series of papers in leading conferences and journals in computer vision, artificial intelligence, and assistive technology, including IROS, ACVR, ICIP, ICARA and Journal of Imaging. His work has been recognized for its contributions to real-world accessibility solutions. Additionally, he has served as a teaching assistant for courses in computer vision and machine learning and has mentored graduate students in their research.

This dissertation was completed under the guidance of Professor Yi Fang at the NYU Embodied AI and Robotics Lab.

Acknowledgements

I would like to express my deepest gratitude to my advisor, Professor Yi Fang, for his unwavering support, invaluable guidance, and encouragement throughout my Ph.D. studies. His profound expertise in deep learning and assistive technology has been instrumental in shaping my research and inspiring new ideas. His patience, motivation, and dedication to academic excellence have provided me with a strong foundation for both my research and professional growth. I am sincerely grateful for his insightful advice, continuous mentorship, and the resources he provided, which have made my research journey both enriching and stimulating. I would also like to sincerely thank the members of my thesis defense committee: Professor Anthony Tzes, Professor Edward Wong, and Professor Chinmay Hegde. I am truly grateful for their time, thoughtful questions, and constructive feedback, all of which have helped me refine and strengthen this dissertation. Furthermore, the members of the NYU Embodied AI and Robotics Lab have contributed immensely to both my professional development and personal growth during my time at New York University. I would like to express my heartfelt thanks to Dr. Hao Huang, Dr. Shuaihang Yuan, Dr. Congcong Wen, and Dr. Lingjing Wang for their insightful feedback and support. Their contributions, particularly in shaping the theoretical components of this dissertation, and their help with the development of visualizations and experimental frameworks, were invaluable to my research. Their perspectives and scholarly rigor have been deeply appreciated. Moreover, I would like to thank all my friends, especially Junchi Feng, Zhongzheng Yuan, and Haoyang Pei, for their encouragement during my Ph.D. studies.

Yu Hao

May 2025

To all the Ph.D. pursuing brave souls

ABSTRACT**Computer Vision for Assistive Technologies: Enhancing Environmental
Perception for the Visually Impaired****by****Yu Hao****Advisor: Prof. Yi Fang, Ph.D.****Submitted in Partial Fulfillment of the Requirements for
the Degree of Doctor of Philosophy (Computer Science)****May 2025**

People with blindness and low vision (pBLV) face significant challenges in daily life due to their reliance on fragmentary auditory, tactile, and proprioceptive cues. they struggle to locate specific target objects, understand complex scenes, and assess potential hazards in dynamic surroundings. They also face challenges in interacting with objects and navigating safely through cluttered indoor and outdoor spaces. Traditional assistive technologies, such as white canes, guide dogs, and GPS-based navigation systems, offer some support but often lack precision,

adaptability, and real-time contextual awareness. These limitations highlight the urgent need for next-generation assistive technologies that can perceive, understand, and act dynamically to enable greater autonomy and safety for pBLV users.

To meet this need, this dissertation introduces an integrated framework that fuses Perception (See), Multi-modal Reasoning (Think), and Embodied Assistive Robotics (Act) into a unified pipeline. At the See level, a lightweight wearable system is developed to support close-range object localization for pBLV users. This system employs deep learning-based detection combined with a resolution-adaptive YOLO model, dynamically adjusting input resolution and improving localization accuracy under diverse real-world conditions. At the Think level, the research proposes multimodal vision-language models, such as ChatMap, that anchor environmental knowledge in structured scene graphs, enabling real-time question answering, semantic scene understanding, and personalized interaction through prompt engineering and spatial reasoning. Finally, at the Act level, the dissertation presents an embodied assistive robot to assist the pBLV to interact with the surrounding environment, which can understand and decompose spoken user instructions into structured robotic actions.

First, the dissertation introduces a close-range object localization framework to help pBLV users identify and reach objects of interest in unfamiliar environments. A lightweight, monocular vision-based wearable system is developed, integrating deep learning-based object detection, a resolution-adaptive YOLO model (RA-YOLO), and real-time trajectory-aware navigation. This system dynamically adjusts to different viewing conditions, enabling accurate object localization and efficient path planning with minimal hardware, thereby enhancing independent mobility at close range.

Second, the research explores multi-modal foundation models for enhanced scene understanding, focusing on enabling pBLV users to build a richer, personalized understanding of their surroundings. Leveraging large vision-language models, the dissertation introduces platforms such as ChatMap, which anchor environmental knowledge into structured scene graphs. Through multimodal fusion and prompt-driven reasoning, these systems support interactive spatial cognition, dynamic question answering, and context-aware scene descriptions, enabling pBLV users to engage with and query their environment naturally.

Third, the dissertation investigates assistive robotics for accessibility, extending intelligent support from perception and understanding to autonomous physical interaction. The proposed model translates spoken commands into structured robot actions, allowing robots to plan and perform complex multi-step object manipulations.

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Chapter 1

Introduction

1.1 Motivation and Research Background

According to 2020 WHO estimates, 295 million people worldwide suffer from moderate to severe visual impairment, while 43.3 million people are blind [101]. Between 1990 and 2020, the number of individuals with moderate to severe visual impairment increased by 91.7%, and the number of people who are blind rose by 50.6% [51]. This trend is predicted to continue, with projections estimating 474 million people with moderate to severe visual impairment and 61 million people experiencing blindness by 2050 [98].

Blindness and low vision pose significant challenges in nearly all activities of daily living, requiring individuals to navigate environments, recognize objects, and interact with the world without full visual capabilities [89]. A fundamental aspect of many daily activities is visual search, which involves actively scanning the environment to locate a specific target among irrelevant distractors [123]. This task is inherently demanding, even for individuals with normal vision, and becomes

significantly more challenging for pBLV [86].

For people with peripheral vision loss, central vision loss, or hemi-field vision loss, reduced field of view makes it difficult to isolate and locate specific objects. Those with blurred vision or nearsightedness struggle to identify objects at a distance, while individuals with color-deficient vision or low contrast sensitivity may find it difficult to distinguish objects from their backgrounds if they share similar colors. Beyond the challenge of identifying an object’s location, approaching and reaching the object safely remains a critical difficulty. pBLV users frequently need continuous guidance, not just initial information about an object’s location, but real-time assistance in navigating toward it while avoiding obstacles [36].

1.2 Challenges in Assistive Technology for pBLV

Assistive technologies designed to support pBLV have made considerable advancements. However, significant challenges remain that hinder their effectiveness, particularly in the ability to locate final destinations or target specific objects in unfamiliar environments [92]. Current deep learning-based object detection models are highly sensitive to environmental factors such as image quality, lighting conditions, and occlusions [52]. These sensitivities often lead to missed or false detections, especially in cluttered or dynamic scenes. Furthermore, monocular camera-based systems struggle to estimate object distances accurately due to the lack of depth perception, making close-range navigation and target acquisition difficult. Traditional GPS-based navigation aids are also ineffective indoors or in urban settings where signal accuracy drops significantly, with errors reaching up to 20 meters [44], further complicating the task of reaching precise locations without

external help.

Beyond recognizing objects, pBLV users must also understand complex scenes and assess potential hazards in dynamic environments. While many object detection models can identify items, they often lack semantic reasoning capabilities [109]. For example, recognizing a traffic light without understanding its current state (red or green) or detecting a crosswalk without assessing its accessibility does not offer sufficient guidance. Furthermore, traditional assistive technologies do not provide risk assessments, such as alerting users to sudden elevation changes, moving obstacles, or temporary hazards like construction zones [57]. This lack of contextual scene understanding leaves users vulnerable when navigating unfamiliar environments.

Finally, pBLV users face substantial challenges in interacting with objects and navigating safely through cluttered indoor and outdoor spaces. While scene understanding provides cognitive awareness, real-world interaction demands precise, physical engagement [1]. Most existing systems lack the capability to assist users with tasks like grasping specific objects, placing them accurately, or safely maneuvering through narrow, obstacle-filled pathways [30]. Robotic systems intended for assistive manipulation often require extensive computational resources and external hardware, making them impractical for daily wearable use. Therefore, developing embodied assistive systems that combine efficient real-time perception, structured task planning, and dynamic manipulation remains a critical, unresolved challenge for enabling true autonomy for pBLV.

1.3 Methods and Contributions

This dissertation aims to advance assistive technology for pBLV by addressing key challenges in object localization, scene understanding, physical interaction, and spatial perception. Despite significant progress in computer vision and deep learning, current assistive technologies often suffer from limited precision, contextual reasoning, and real-time adaptability. To bridge these gaps, this research proposes a unified framework grounded in deep learning and multi-modal AI. The dissertation is structured into four major parts, each tackling a distinct but complementary aspect of the problem.

First, the dissertation presents methods for *Close-Range Object Localization for pBLV*. Detecting and approaching objects in unfamiliar environments is challenging due to variable image quality, occlusions, and varying object distances. This work introduces a deep learning-powered, monocular vision-based wearable system that enables real-time object detection, localization, and trajectory estimation. It begins by evaluating the impact of image resolution and distance on detection performance using YOLO-based models. To improve robustness, a resolution-adaptive YOLO (RA-YOLO) object detection model is proposed. The system continuously updates the 3D location of target objects relative to the user’s position using trajectory-aware algorithms, providing real-time path planning and correction for enhanced navigation accuracy.

Second, to go beyond object recognition and enable real-world environmental interaction, the dissertation explores *Multi-Modal Foundation Models for Enhanced Scene Understanding*. This work introduces the wearable AI frameworks Chatmap that integrate pre-trained vision-language foundation models with spatial reasoning and prompt engineering. These models enable context-aware scene understanding,

real-time question answering, and hazard assessment through multi-modal inputs. They are designed to support natural user interaction and dynamic environmental interpretation, empowering pBLV individuals with personalized, intelligent assistance in both indoor and outdoor settings.

Third, the dissertation investigates *Assistive Robotics for Accessibility*, expanding assistive capabilities into task execution and embodied reasoning. The Chain of Robotic Actions Reasoning for Autonomous Robotic Arm Manipulation (CORA) model is introduced to support multi-step robotic manipulation through high-level action planning and sequential reasoning. Additionally, Embodied Spatial Reasoning for Enhanced Situated Question Answering in 3D Scenes (EMBOSR) is proposed to support situated question answering in 3D environments. These contributions demonstrate how intelligent robotic systems can provide assistive support for object interaction, spatial inference, and physical accessibility in real-world scenarios.

Together, these research directions contribute to a comprehensive computer vision-powered framework for enhancing accessibility, safety, and independence for pBLV. The proposed methods have been validated through extensive experiments and system implementations, demonstrating their practical impact in real-world assistive scenarios.

Chapter 2

See: Close-Range Object Localization for pBLV

2.1 Detect and Approach: Close-Range Navigation Support for pBLV

pBLV experience significant challenges when locating final destinations or targeting specific objects in unfamiliar environments. Furthermore, besides initially locating and orienting oneself to a target object, approaching the final target from one's present position is often frustrating and challenging, especially when one drifts away from the initial planned path to avoid obstacles. In this section, we aim to develop a novel wearable navigation solution to provide real-time guidance for a user to approach a target object of interest efficiently and effectively in unfamiliar environments [52]. Our system contains two key visual computing functions: initial target object localization in 3D and continuous estimation of the user's trajectory, both based on the 2D video captured by a low-cost monocular camera mounted on